

Causal Inference Workshop

Week 11 - Field Experiments in Sustainable Development and **wrap-up!**

Causal Inference Workshop

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Workshop outline

A. Causal inference fundamentals

- Modeling assumptions matter too
- Conceptual framework (potential outcomes framework)

B. Design stage: common identification strategies

- IV + RDD [coding]
- DiD, DiDiD, Event Studies, New TWFE Lit [coding]
- Synthetic Control / Synthetic DiD [coding]

C. Analysis stage: strengthening inferences

- Limitations of identification strategies, pre-estimation steps
- Estimation [controls] and post-estimation steps [supporting assumptions]

D. Other topics in causal inference and sustainable development

- Inference (randomization inference, bootstrapping, etc.)
- Fixed effects, climate regressions, remote sensing data
- Field experiments in sustainable development and wrap up!

Outline

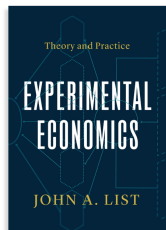
Workshop outline

Experiments

Recap of the Causal Inference Workshop

Presenting your paper

A Crash Course on John List's Experimental Economics



1. Introduction
2. A Primer on economic experiments
3. Internal validity
4. Statistical conclusion
5. Optimal experiment
6. Randomization techniques
7. Heterogeneity and causal moderation
8. Mediation
9. Experiments and longitudinal elements
10. Within-subject experimental designs
11. SUTVA
12. Observability
13. Complete compliance
14. Statistical independence
15. Building confidence in experimental results
16. Generalizability and scaling
17. The ethics of economic experiments
18. Pre-treatment administrative responsibilities
19. Optimal use of incentives in economic experiments
20. Epilogue: the written road to scientific knowledge diffusion

Internal and External Validity (Ch1)

- **Experiment Problem 1 (EP1):** Quantifying economic fundamentals, measuring treatment effects, and identifying relevant mediators and moderators in an ethically responsible manner.
- **Experiment Problem 2 (EP2):** Predicting whether the causal impacts of treatments implemented in one environment transfer to other environments, whether spatially, temporally, or scale differentiated.

Experimental Types (Ch2)

- **Laboratory experiment:** subjects are typically college students; has an abstract framing and an imposed set of rules.
- **Artefactual field experiment:** subjects drawn from the market of interest; has an abstract framing and an imposed set of rules.
- **Framed field experiment:** subjects drawn from the market of interest; adds naturalness to the situation.
- **Natural field experiment:** unknowingly, subjects take part in an experiment; combines randomization and realism.

Assumptions to Recover ATE (Ch3)

- **Assumption 3.1 Stable Unit Treatment Value Assumption (SUTVA):** For all units $i \in \mathcal{J}$, for all $\mathbf{d} \in \mathcal{D}^N$, $Y_i(\mathbf{d}) = Y_i(d)$.
 1. Any unit's potential outcomes do not vary with the treatment undertaken by any other unit.
 2. There is no variation in the form or version of the treatment that leads to different potential outcomes.
- **Assumption 3.2 Observability:** For all units $i \in \mathcal{J}$, we have $\mathbb{P}[R_i = 1] = 1$, and the researcher observes $(Y_i, D_i, Z_i, \mathbf{X}_i)$.
- **Assumption 3.3 Complete Compliance:** For all units $i \in \mathcal{J}$, $\mathbb{P}[D_i = Z_i] = 1$.
- **Statistical Independence:** For all units $i \in \mathcal{J}$, $D_i \perp\!\!\!\perp \{Y_i(d)\}_{d \in \mathcal{D}}$.

Estimation and Hypothesis Testing (Ch 4)

Estimator for ATE

- Difference-in-means
- Difference-in-Differences
 - Balanced Panel Assumption

Estimation and Hypothesis Testing (Ch 4)

- Single hypothesis testing and statistical power.
- Multiple hypothesis testing. Well-defined families of hypotheses rather than simple individual tests.
 - Multiple outcomes of interest
 - The effect of a treatment may be heterogeneous across subgroups
 - Multiple treatments of interest

Natural analog to significance level: Family-Wise Error Rate

- Bonferroni Correction
- Holm Stepdown Correction
- List et al. FWER Correction

Optimal Experimental Design: Maximize Your Power! (Ch 5)

To derive the optimal treatment level (in the case of multi-valued treatment), and the size of each group, need to consider:

- desired significance level, desired statistical power, effect size
- experimental budget and treatment cost
- how the data will be analyzed
 - unit of assignment (design with clustered assignment)
 - number of distinct outcomes of interest (design for multiple hypothesis testing)
- available pre-treatment covariates

Optimal Experimental Design: Maximize Your Power! (Ch 5)

Simplest case: binary treatment D_i , n_1 treated and n_0 control.

- Simple because of binary treatment, no clustered design, single hypothesis testing, not considering pre-treatment covariates
- Significance level α , statistical power β , true ATE $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$.
- Treated unit cost c_1 , control unit cost c_0 , budget m .
- $\sigma_0^2 = \mathbb{V}(Y_i(0))$, $\sigma_1^2 = \mathbb{V}(Y_i(1))$.

Goal is to choose n_1 and n_0 to maximize β given α , τ , σ_1 , σ_0 while complying with the budget. (Maximize β given τ is equivalent to minimizing detectable effect size given β .)

Optimal Experimental Design: Maximize Your Power! (Ch 5)

Goal is to choose n_1 and n_0 to maximize β given $\alpha, \tau, \sigma_1, \sigma_0$ while complying with the budget.

$$\begin{aligned} \max_{n_1, n_0} \quad & \beta = \Phi\left(-\Phi^{-1}(1 - \alpha/2) + \frac{\tau}{\sqrt{\sigma_1^2/n_1 + \sigma_0^2/n_0}}\right) \\ \text{s.t.} \quad & c_1 n_1 + c_0 n_0 = m \end{aligned}$$

- We may not know τ, σ_1, σ_0
 - Learn about them from pilot and literature.
 - Calculate the optimal (n_1, n_0) and the corresponding maximized β for different values. If assume $\sigma_1 = \sigma_0 = \sigma$, then what matters is the ratio τ/σ alone.
- One observation: $\frac{n_1^*}{n_0^*} = \sqrt{\frac{c_0}{c_1} \frac{\sigma_1}{\sigma_0}}$. When $c_1 = c_0$ and $\sigma_1 = \sigma_0$, we have $n_1^* = n_0^*$.

Randomization Techniques (Ch 6)

- Bernoulli Trials
 - Random draw $D_i \sim \text{Bernoulli}(p)$ for each unit independently.
 - Work for large sample size.
- Complete Randomized Experiments (CRE)
 - Fix N_t and N_c before randomization
 - $\mathbb{P}(\mathbf{D} \mid \mathbf{Y}(\mathbf{0}), \mathbf{Y}(\mathbf{1}), \mathbf{X}) = \frac{N_t!N_c!}{N!}$
 - Randomization is ex ante unbiased, but any particular draw can still be unlucky.
 - Even though $\mathbb{E}[\hat{\tau}] = \tau$, the estimate from one particular draw may have a large deviation from τ .

Randomization Techniques (Ch 6)

- Block/Stratified Experiments
 - Partition units into strata on discrete covariates, then run a CRE within each stratum.
 - Advantages
 - Exact balance on stratifying variables by construction
 - Reduces variance $\rightarrow$ higher power (when strata predict outcomes)
 - Enables credible CATE estimates (not data mining)
 - Improves external validity (can extrapolate CATEs to target population)
 - Suitable with strongly predictive, low-dimensional, discrete covariates.

Randomization Techniques (Ch 6)

- Matched-Pairs Designs. Example: Bruhn et al. (2016)

Step 1 – Outer strata: partition 892 schools into 12 strata using discrete variables (financial institutions per capita $\times$ Brazilian state)

Step 2 – Within-stratum matching (“greedy” algorithm):

1. For each school, compute Mahalanobis distance to every other school in the stratum using 6 continuous covariates: GDP per capita, savings per capita, # students, # teachers, dropout rate, graduation rate
2. Find the pair with the smallest pairwise distance $\rightarrow$ match them
3. Remove that pair, repeat until all schools are matched

Step 3 – Within-pair CRE: flip a coin to assign treatment/control within each pair

Randomization Techniques (Ch 6)

- Rerandomization
 - Randomization, check balance, and if it fails, try again. Do it formally with a pre-specified rule.
 - A rerandomization plan is a function of $\mathbb{D}$ and $\mathbb{X}$ that defines admissible randomizations ex ante. [▶ app: rerandom](#)
 - Ex ante, rerandomization is equivalent to regression-adjusting for balance covariates, but it is done at the design stage; prevents inference issues
- Design-conscious Inference
 - The standard two-sample t-test applies to CRE and are often too conservative for other randomization techniques.
 - Randomization assignment mechanisms affect how to do estimation and statistical inference.

Moderator (Ch 7)

- Descriptive moderation analysis
 - Parameter of interest CATE: $\tau(x) \equiv \mathbb{E}[\tau_i \mid X_i = x] = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$.
 - Identifying Assumption $\{Y_i(1, X_i), Y_i(0, X_i)\} \perp\!\!\!\perp D_i \mid X_i$.
 - Conventional approach suitable for a quite small group of discrete X.
 - Modern approach: causal forests, adapted from the classification and regression tree approach. Examples: Holz et al. (2023)

Moderator (Ch 7)

- Causal moderation analysis
 - Causal parameter of interest, the average difference in treatment effect when X_i is x vs x' , Average Treatment Moderation Effect:

$$ATME(x, x') \equiv \mathbb{E}[(Y(1, x) - Y(0, x)) - (Y(1, x') - Y(0, x'))]$$

- Assumptions needed:
 - SUTVA-Moderator: potential outcomes do not vary with the treatments or causal moderators assigned to any other unit.
 - Statistical independence of treatment assignment and pre-treatment of moderator:
 $\{Y_i(d, x), X_i\}_{d \in \mathcal{D}, x \in \mathcal{X}} \perp\!\!\!\perp D_i$
 - Statistical independence of moderator: $\{Y_i(d, x)\}_{d \in \mathcal{D}, x \in \mathcal{X}} \perp\!\!\!\perp X_i$
- Ideally random X_i to have those independence assumptions to hold.
- Under these assumptions, $ATME(x, x') = \tau(x) - \tau(x')$.

Moderator (Ch 7)

To sum up

- Moderator cannot be moved by D_i .
- If you plan to do descriptive moderation analysis, do stratified randomization.
- If you plan to do causal moderation analysis, randomize X_i
- Avoid ex-post moderation analysis that is not planned in the design stage (avoid cherry picking)

Mediator (Ch 8)

- Mediators are moved by treatment. Mediation analysis: what is the causal pathway from D to Y ?
- Complete mediation: the entire effect of D on Y is transmitted through one or more mediator variables.
- Partial mediation, D has both a direct and an indirect effect on Y .
- Potential outcome $Y_i(M(d), d)$ as the outcome arising when $D_i = d$.
- Parameter of interest
 - Average direct effect $ADE(d) \equiv \mathbb{E}[Y_i(M_i(d), 1) - Y_i(M_i(d), 0)]$
Average indirect effect $AIE(d) \equiv \mathbb{E}[Y_i(M_i(1), d) - Y_i(M_i(0), d)]$
- $ATE = \mathbb{E}[Y_i(M_i(1), 1) - Y_i(M_i(0), 0)] = ADE(1) + AIE(0) = ADE(0) + AIE(1).$

Mediator (Ch8)

- Parameter of interest
 - Average direct effect $ADE(d) \equiv \mathbb{E}[Y_i(M_i(d), 1) - Y_i(M_i(d), 0)]$
Average indirect effect $AIE(d) \equiv \mathbb{E}[Y_i(M_i(1), d) - Y_i(M_i(0), d)]$
- $ATE = \mathbb{E}[Y_i(M_i(1), 1) - Y_i(M_i(0), 0)] = ADE(1) + AIE(0) = ADE(0) + AIE(1).$

Additive Separability Assumption: D and M have additively separable contributions to Y ; that is $Y_i(m, d) = f_i(m) + g_i(d)$. Then,

- $ADE(d) = \mathbb{E}[(f_i(M_i(d)) + g_i(1)) - (f_i(M_i(d)) + g_i(0))] = \mathbb{E}[g_i(1) - g_i(0)]$
- $AIE(d) = \mathbb{E}[(f_i(M_i(1)) + g_i(d)) - (f_i(M_i(0)) + g_i(d))] = \mathbb{E}[f_i(M_i(1)) - f_i(M_i(0))]$

Thus, $ADE(1) = ADE(0) = ADE$, $AIE(1) = AIE(0) = AIE$, and $ATE = ADE + AIE$.

Mediator (Ch8)

Paired Design: two parallel experiments

- Experiment A: the researcher conducts a standard experiment, randomly assigning D
→ estimate $ATE \equiv \mathbb{E}[Y_i(M_i(1), 1) - Y_i(M_i(0), 0)] = ADE + AIE$
- Experiment B: the researcher utilizes a full factorial design in $D \times M$
→ estimate all the distinct ACDEs. Average controlled direct effects $ACDE(m) \equiv \mathbb{E}[Y_i(m, 1) - Y_i(m, 0)]$. Under the Additive Separability Assumption, the researcher is expected to find that $ACDE(m)$ does not depend on m :
 - $ACDE(m) = \mathbb{E}[(f(m) + g(1)) - (f(m) + g(0))] = \mathbb{E}[g(1) - g(0)] = ADE$.
- Combine the results of the two experiments together
→ $AIE = ATE - ADE$. ATE is from Experiment A, while ADE is from Experiment B.

Mediator (Ch 8)

Other approaches to learn about mediation: separate randomization and cross-over design. Separate randomization and its problem

- Conduct a first experiment whereby D is randomized and measure outcome M
- Followed by a second experiment that randomizes M and measures outcome Y
- The naturally prevailing levels of the mediator in the first experiment could differ from the assigned level of mediator in the second experiment.

Exhibit 8.4: A Simple Example of Separate Randomization of M and D

Population Proportion	Potential Mediators and Outcomes				Treatment Effect on Mediator $M_i(1) - M_i(0)$	Mediator Effect on Outcome $Y_i(t, 1) - Y_i(t, 0)$	Causal Mediation Effect $Y_i(t, M_i(1)) - Y_i(t, M_i(0))$
	$M_i(1)$	$M_i(0)$	$Y_i(t, 1)$	$Y_i(t, 0)$			
0.3	1	0	0	1	1	-1	-1
0.3	0	0	1	0	0	1	0
0.1	0	1	0	1	-1	-1	1
0.3	1	1	1	0	0	1	0
Average	0.6	0.4	0.6	0.4	0.2	0.2	-0.2

More Complex Designs

- Ch 9 Experiments with longitudinal elements
 - Longitudinal elements in treatment: treatment duration and timing, multiple exposures
 - Longitudinal elements in outcomes: pre-treatment outcomes, long-run outcomes.
- Ch 10 Within-subject design
 - Each unit participates in treatment and control in different periods.
 - Need balanced panel, temporal stability, and causal transience assumptions (no anticipation or carry-over).

Generalization and Scaling (Ch 16)

- (Horizontal) Generalizability: taking the insights from one experiment and applying it to other settings with different environments, tasks, or contexts.
- Scaling / Vertical Generalizability: taking the results from one experiment and understanding how the results from that experiment will change as the number of participants, or scale, increases.

Generalization and Scaling (Ch 16)

An experiment identifies:

$$\mathbb{E}[Y_i(1) - Y_i(0) \mid P_i = 1, E, A, X_i]$$

- **Participation (P):** We observe effects for participants ($P_i = 1$), but want to learn the ATE of the total population including non-participants ($P_i = 0$).
- **Environment (E):** We estimate effects in the study environment, but may want the ATE in a different context or setting.
- **Experimental scrutiny (A):** Subjects may behave differently under observation; we may want the ATE without such scrutiny.
- **Covariates (X_i):** Results apply to the observed covariate distribution, but we may want the ATE at other values or in other populations.

Pre-treatment Administrative Responsibilities (Ch 18)

Internal Board Review

- Do not be intimidated by this process. Human Research Protection Office at CU is very supportive.
- Typical process: create and submit a protocol in Rascal (need a faculty PI); get feedback; check with the assigned reviewer with questions and set up a meeting with them if necessary; revise and submit again; get approved; update if necessary (e.g., add IRB approval of your coauthor from their institution)

Pre-treatment Administrative Responsibilities (Ch 18)

Register your study in the AEA RCT Registry with a pre-analysis plan (PAP)

- Basic registration can be completed in 5–10 minutes.
- A pre-analysis plan (PAP) is extremely valuable. It does not mean that you are 100% bound by it.
 - A PAP should include details on data cleaning, variable construction, and econometric analysis (e.g., main specifications and heterogeneity analysis). Writing it forces you to think if your experimental design can support your intended analyses.
 - In your paper, clearly indicate any deviations from the PAP and provide justification. Additional details can be included in the appendix.

Write an Experimental Paper (Ch 20)

Area	Item number	Checklist item
Title and Abstract		
	1	Identification as a lab or field experiment in the title
	2	Structured summary of experimental design, methods, results, and conclusions
Introduction		
Backdrop, Evidence & Conclusion	3	Motivation, describing core concept and knowledge gaps
	4	Outlining results and establishing internal validity
	5	Examining external validity
	6	Literature Overview
Theory		
	7	Predictions and how they are tested
Experimental Methods		
Participants	8	Eligibility criteria for participants
Situation	9	Recruitment approach used
Situation	10	Settings and locations where the data were collected
Experimental approach	11	Randomization method
Experimental approach	12	Sample size determination and power tests
Experimental approach	13	Outcomes measured
Experimental approach	14	Between- or within-design
Experimental approach	15	Mediators and moderators
Experimental approach	16	Ethical considerations
Statistical Approach	17	Methods for data analysis and balance table
Experimental Results and Discussion		
Presentation	18	Registered (PAP) versus non-pre-registered (PAP) insights
Presentation	19	PREP Flowchart
Epilogue	20	Paths forward
Other Materials	21	Pre-registration, PAP, conflicts of interest and appendices

Other Chapters

- Ch 11 SUTVA
- Ch 12 Observability
- Ch 13 Complete compliance
- Ch 14 Statistical independence
- Ch 15 Building confidence in experimental results
- Ch 17 Ethics
- Ch 19 Optimal Use of Incentives

Last Messages

- The book has a lot of materials related to more complex design, do check them out.
 - Power calculation for clustered design (Ch 5)
 - Multiple treatment arms and factorial designs (Ch 6)
- Pilot is important. Do make the most use of it.
- Spend enough time on designing and collecting data for designing (for example, collecting X for stratified randomization); the effort will pay off. Careful design will save you from a lot of work in estimation.

Outline

Workshop outline

Experiments

Recap of the Causal Inference Workshop

Presenting your paper

Causal inference using observational data

1. Framework: A counterfactual approach to causality

- Potential outcomes framework *week 2*
- Causal graph (DAG) framework *week 2*

2. Design stage

- IV *week 3* [coding, week 4]
- RD *week 3* [coding, week 4]
- DiD, DiDiD, TWFE *week 5* [coding, week 6]
- SCM *week 7* [coding, week 7]

3. Analysis stage

- Pre-estimation: restructuring data *week 8*
- Estimation: regression controls, inference *week 8 & 9* [coding, week 9]
- Post-estimation: Checking/supporting assumptions *week 8*

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1. Framework
2. Design stage
3. Analysis stage

Presenting your paper

What am I trying to estimate?

- We want to estimate a TE, defined by potential outcomes
 - We can only measure observed outcomes (fundamental problem of causal inference)
- We estimate a TE using relationship between observed outcomes and potential outcomes

individual treatment effects (TEs)	$Y_i^1 - Y_i^0 \forall i$	<i>ideally estimate; unknowable</i>
average treatment effect (ATE)	$\mathbb{E}[Y_i^1 - Y_i^0]$	<i>reasonably estimate; unknowable, but can be estimated</i>
average treatment effect on the treated (ATT)	$\mathbb{E}[Y_i^1 - Y_i^0 D_i = 1]$	<i>reasonably estimate; unknowable, but can be estimated</i>
difference in average observed outcomes	$\mathbb{E}[Y_i D_i = 1] - \mathbb{E}[Y_i D_i = 0]$	<i>what we can estimate</i>

Independence assumptions identify TEs and give unbiased estimators

- We can only compute the difference in average *observed* outcomes
- We can recover an **unbiased** estimator of a causal effect iff an **identifying/independence assumption** holds:
 - if IA holds $((Y_i^0, Y_i^1) \perp\!\!\!\perp D_i) \rightarrow$ estimate ATT
 - if ~~IA~~, but CIA $((Y_i^0, Y_i^1) \perp\!\!\!\perp D_i | X_i) \rightarrow$ can estimate ATT in each stratum (and then combine)
 - if ~~CIA~~, need relevant exogenous source of variation in D_i (e.g., $(Y_i(d, z), D_i(z)) \perp\!\!\!\perp Z_i$, $Y_i(d, z) = Y_i(d)$, $D_i(1) \geq D_i(0)$) $\rightarrow$ estimate LATE
- Estimate of $\hat{\beta}$ is never the true effect, only one realization of an unbiased *distribution*

Expressing TE as linear regression

$$Y_i = Y_i^0 + (Y_i^1 - Y_i^0)D_i = \dots = \alpha + \beta D_i + u_i$$

Which means that:

- $\hat{\beta}_{OLS}$ is unbiased for the ATT iff:
 - there is no selection bias (identification problem; independence)
 - e is uncorrelated with D (regression problem, endogeneity)

In observational studies:

- Excluding a confounder creates bias, so we must adjust for all confounders
- With all confounders adjusted for, we have an unbiased estimator of an average TE
- As we can rarely be certain to have measured all confounders, turn to **identification strategies** that rely on other assumptions

Outline

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Recap of the Causal Inference Workshop

1. Framework
2. Design stage
3. Analysis stage

Presenting your paper

Common Identification Methods

1. **RCT** or 'natural' randomization of treatment
2. **IV, RD**: *If there may be selection based on unobservables, we use an instrument or discontinuity that induces exogenous variation in treatment status.*
3. **DiD, DiDiD, Event Studies, SCM**: *If we have repeated observations and want to estimate the effects of an event, we use research designs that assume or construct parallel trends and only time-invariant confounds.*
 - Event-studies: All units get treated; we identify effects from *within* variation only
 - DiD: Some units never get treated; we identify effects from *within* and *between* variation
 - SCM: Combine control units to construct synthetic unit (counterfactual)

→ We reviewed the assumed DGP, identifying assumptions, the estimated, the estimator used, and best practices for each!

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All Other Things (Besides Identification!)

1. **Pre-estimation:** Restructuring data, so we rely on functional form of $f(X)$ less
2. **Estimation:** What controls do we include?
 - *Required/forbidden controls:* **adjust** for all confounding paths, but **do not** adjust for post-treatment variables that may be affected by treatment
 - *Optional good/bad controls:* can adjust for pre-treatment covariates that are strong determinants of Y (may increase efficiency of $\hat{\beta}$); do not adjust for determinants of D (may decrease precision)
3. **Post-estimation:** *"Falsification tests set you up for failure: do them."* - Doug
 - *Falsification tests:* Show that specification doesn't find an effect when one should not exist
 - *Diagnosis tests of modeling assumptions:* Easy to test basic assumptions of CLRM (non-linearity, homoskedasticity of errors, leverage of observations)

Outline

Workshop outline

Experiments

Recap of the Causal Inference Workshop

Presenting your paper

Characterizing the empirical strategy

Your presentation/paper should contain – to some degree, explicitly:

1. **Research question**

What causal effect of interest are we trying to estimate?

2. **Ideal experiment**

What ideal experiment would capture the causal effect?

3. **Identification strategy**

How are the observational data used to make comparisons that approximate such an experiment?

4. **Estimation method** (including assumptions made when constructing standard errors)

5. **Falsification tests** that support the identifying assumptions.

Putting the paper in perspective

In addition to the paper's empirical strategy, one may want to discuss:

- **Contributions to the literature**
- **Methodological contributions**
- **Internal validity** of the statistical analysis
 - *Are the identifying assumptions plausible? Could there be measurement error? Are there unexplained results?*
- **External validity** of the statistical analysis
 - *w.r.t. to policy:* Is there a gap between policy questions and the analyses performed?
 - *w.r.t. to the literature:* How does the paper account for its results compared to other results in the literature?
 - *w.r.t. to other settings:* Are the results generalizable to other populations and settings?

Structuring presentation

- Introduction [forecast]: ~10 minutes (5-7 for 30-min talk)
- Middle [do]: ~30 minutes (12-15 min for 30-min talk)
- End [takeaways]: ~5-10 minutes (<5 min for 30-min talk)

Structuring presentation

- Introduction [forecast]: ~10 minutes (5-7 for 30-min talk)
 - Topic and research question (should get to it relatively quickly; general vs. specific research question)
 - Answer to question (findings)
 - Relationship/contribution to literature
 - Importance - why should we listen to you?
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 - Takeaways you want audience to remember

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 - Practice polite but firm things to say for questions and to control your own timing

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 - Practice polite but firm things to say for questions and to control your own timing
- Delivery
 - Practice talking about research
 - Practice out loud

Questions? Comments?

Thank you!

References

Heavily based on Claire Palandri's 2022 version and Anna Papp's 2024 version of the Causal Inference Workshop, and course materials of Experimental Economics Spring 2026 offered by Robert Metcalfe.

- Bruhn, Miriam, Luciana de Souza Leão, Arianna Legovini, Rogelio Marchetti, and Bilal Zia. 2016. "The Impact of High School Financial Education: Evidence from a Large-Scale Evaluation in Brazil." *American Economic Journal: Applied Economics* 8 (4): 256–95. <https://doi.org/10.1257/app.20150149>.
- Holz, Justin E, John A List, Alejandro Zentner, Marvin Cardoza, and Joaquin E Zentner. 2023. "The \$100 million nudge: Increasing tax compliance of firms using a natural field experiment." *Journal of Public Economics* 218:104779.

4-Step Protocol of Rerandomization

1. Specify a balance criterion ex ante (e.g., all p-values > 0.05 , or Mahalanobis distance below a threshold)
2. Randomize units to treatment and control (e.g., via CRE)
3. Test for balance. Satisfied $\rightarrow$ proceed; Dissatisfied $\rightarrow$ back to step 2
4. Conduct experiment and adjust inference to account for the rerandomization

◀ rerandom